AHMAD HASAN

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SUMMARY

Recent Master's graduate in Applied Computer Science with a Microprogram in Applied AI. Hands-on experience building deep learning models (CNNs, Transformers, BERT), full-stack applications, and backend services. Strong fundamentals in Python, Java, and C++ with practical exposure to TensorFlow, PyTorch, and cloud platforms.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, SQL, HTML/CSS

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, Keras, Hugging Face Transformers, PEFT (LoRA), NLTK, SpeechBrain, OpenCV, BERT, CNNs, Transformers

Data & Analytics: Pandas, NumPy, SQL, Power BI, Tableau, ETL, Dimensional Modeling

Cloud & DevOps: Google Cloud Platform (Digital Leader Certified), Git, GitHub, CI/CD

Backend & Web: Spring Boot, J2EE, Microservices, MERN Stack, REST APIs, SOAP

Tools & Methods: Jupyter, VS Code, IntelliJ, Jira, Agile/Scrum, Code Refactoring, JUnit

EDUCATION

Microprogram in Applied Artificial Intelligence | Concordia University, Montreal, QC *2025 – 2026*
GPA: 4.15/4.30 | Coursework: Applied Machine Learning, Applied Deep Learning

Master of Applied Computer Science (CO-OP) | Concordia University, Montreal, QC *2024 – 2026*
GPA: 3.89/4.30 | Coursework: Applied AI, Statistical NLP, Conversational AI, Distributed Systems, Algorithm Design

Bachelor of Science in Computer Science | Ghulam Ishaq Khan Institute (GIKI), Pakistan *2019 – 2023*
Dean's Honor Roll (Spring 2022, Fall 2022, Spring 2023)

PROFESSIONAL EXPERIENCE

Associate Consultant, Data Analytics | Systems Limited *Aug 2023 – Mar 2024*

- Developed and maintained interactive Tableau and Power BI dashboards for an international client, translating raw business data into reporting assets used by stakeholders across the organization.
- Wrote and optimized SQL queries against multi-million row datasets, reducing query runtimes and ensuring metrics displayed in dashboards reflected validated source-of-truth data.
- Applied data engineering practices including ETL pipelines, dimensional modeling, and star schema design to structure data warehouses for downstream analysis.
- Collaborated with cross-functional client teams to translate ambiguous reporting questions into well-defined metric definitions and reusable data products; earned Google Cloud Digital Leader certification.

Courseware Developer | McGill University *Jun 2024 – Present*

- Led collaboration with external vendor (D2L) to deliver a custom student registration platform, gathered requirements across McGill IT, finance, and program teams, drove change requests, built front-end components, and led client-side testing across registration cycles.
- Automated student registration and grading workflows by writing Python scripts to clean and process Excel reports and generate personalized registration links based on each candidate's prior level and category.
- Built interactive learning modules in Articulate Storyline for Dialogue McGill's Language Training Program (600+ healthcare professionals), iterating on features based on student feedback to maintain 95%+ user satisfaction.

Backend Software Developer Intern | PostEx *Jun 2022 – Aug 2022*

- Completed structured training in backend fundamentals: microservices architecture, design patterns, multithreading, servlets, REST and SOAP APIs, and Java enterprise frameworks (Spring Boot, J2EE).
- Built a Java desktop scientific calculator as a capstone exercise applying multithreading, design patterns, and a JavaFX front-end; packaged as a standalone runnable application.
- Contributed backend microservices to the company's internal platform, applying the OOP and Spring Boot patterns covered during training.

SELECTED PROJECTS

LoRA vs Feature Extraction for Scientific QA — *PyTorch, Hugging Face Transformers, PEFT*

- Compared parameter-efficient fine-tuning (LoRA) against frozen-backbone feature extraction on DistilGPT-2 (~82M params) for the SciQ dataset under strict compute constraints.
- Ran three progressive experiments — conservative LoRA baseline, improved LoRA configuration (including a real decoding bug fix uncovered during evaluation), and a frozen-backbone control — measured on Exact Match, Token F1, and Perplexity.

Human vs. AI Text Classification (COLING MGT Workshop) — *Python, Scikit-learn, XGBoost, BERT*

- Developed three classification models (SVM, XGBoost, fine-tuned BERT) to distinguish human-written and AI-generated text.
- Engineered linguistic features including word length, sentence length, and perplexity; compared classical ML against transformer-based approaches.
- Presented findings via research poster covering feature selection, model effectiveness, and detection trade-offs.

IEEE-CIS Fraud Detection — Scalable Classical ML — *Python, scikit-learn, pandas*

- Benchmarked a scalable linear SVM (trained via stochastic gradient descent) against a tuned Decision Tree on the IEEE-CIS fraud dataset, an industry-scale tabular classification problem.
- Performed structured hyperparameter tuning, stratified train/validation splits to preserve class imbalance, and head-to-head model comparison on AUC and F1; documented trade-offs between scalability, interpretability, and predictive performance.

Pure Music – Vocal Isolation with Dual-Path Transformer — *PyTorch, SpeechBrain, MUSDB18*

- Designed a Dual-Path Transformer model for music source separation on the MUSDB18 dataset, inspired by SpeechBrain's modular architecture.
- Built a custom training pipeline with dynamic chunking to process long-form audio efficiently under GPU memory constraints.
- Ran controlled experiments across 2-stem and 4-stem configurations, evaluating separation quality using SDR, SIR, and SAR metrics and documenting trade-offs.

Image Retrieval with Contrastive Learning — *PyTorch, torchvision*

- Built a triplet-network image retrieval system and studied how sampling strategy impacts retrieval quality.
- Ran controlled experiments comparing random triplet selection against semi-hard mining, evaluated using Recall@K and qualitative nearest-neighbor inspection.

Career Craft – AI-Powered Career Platform — *MERN Stack, Python, Web Scraping*

- Built a full-stack MERN web app connecting students and recruiters with ranked job-to-candidate matching.
- Integrated an ML model to predict career paths from user profiles and surface missing skills as personalized recommendations.
- Developed web scrapers to aggregate job postings from multiple sources into a centralized listing.

CERTIFICATIONS & ACHIEVEMENTS

- **Certification:** Google Cloud Digital Leader (valid through 2026)
- **Awards:** Dean's Honor Roll for academic distinction (Spring 2022, Fall 2022, Spring 2023)
- **Languages:** English (Fluent), French (Beginner), Urdu (Native)